

Rigging the Foundation: Manipulating Pre-training for Advanced Membership Inference Attacks

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Abstract—The significant advances in computing power have led to a surge in model complexity. Training such models today increasingly relies on *transfer learning*, where models are pre-trained on large datasets and later fine-tuned for different domains, allowing the knowledge in the pre-trained model to be effectively reused and customized for these specific domains. However, such a learning paradigm also opens new attack surfaces on the fine-tuned model. Particularly, a privacy risk never studied before is the threat posed by the adversary affecting the pre-training process to the downstream user’s private data for fine-tuning the model: A manipulated pre-trained model can render its fine-tuned version vulnerable to privacy attacks, such as *membership inference attacks* (MIAs) where the presence of a given sample in the fine-tuning dataset can be determined by querying the vulnerable model.

A unique challenge in understanding this privacy risk is how to amplify the membership leakage while ensuring the performance of the fine-tuned model. To address this challenge, we introduce a new technique – *active robustness overfitting* (ARO). This approach actively induces *robustness overfitting* during pre-training, which amplifies membership leakage in the downstream task without affecting its accuracy, while also maintaining the stealthiness of the attack. Our extensive evaluations across various datasets and diverse MIA scenarios demonstrate that our methods can effectively amplify membership leakage while preserving satisfactory downstream test accuracy, which contributes to a better understanding of the privacy risk introduced by transfer learning.

1. Introduction

Recent years have witnessed an exponential growth in applications of *deep neural networks* (DNNs) to various domains. With significant advances in computing power, DNNs have become more complicated, both deeper [49] and wider [39] to enhance their performance. These large-scale models, however, require a significant amount of resources to train, including large datasets and substantial computational power. Serving this purpose today is *transfer learning* [44], which pre-trains a model (called *pre-trained model*) on a large dataset, and then shares the model with

different *downstream tasks*, typically through *fine-tuning* the model (called *fine-tuned model* or *downstream model*) on a much smaller dataset for each task [66]. The underlying idea of this approach is to leverage rich, generalized representations from the pre-trained model to augment the model’s performance on the downstream tasks. With its significant contribution to the increasingly wide application of DNNs, this technique, however, also opens up new avenues for attacks on fine-tuned models and their learning process [64]. Particularly, never studied before is a unique privacy risk during transfer learning: the adversary can utilize a carefully crafted or contaminated pre-trained model to help infer sensitive information from its fine-tuned model. A prominent example of such attacks is membership inference attack (MIA) [53], in which one tries to determine whether a given sample is indeed present in the training set of a target model.

New threat and challenges. Although MIAs on DNNs have been extensively studied [7], [22], [53], prior research primarily focuses on the inference threats to upstream [21] or downstream [50] task in isolation. The potential to attack the whole transfer learning process through manipulating the pre-trained model to enhance an MIA on the fine-tuned model is under-explored. Such an attack, however, faces a unique challenge: a pre-trained model would not be used to serve a downstream task if it is not well generalized; starting from a well-generalized model, a downstream model trained on fine-tuning data is known to be much less likely to overfit these data (see [64]), particularly when the fine-tuned model achieves a high accuracy as expected; this renders an MIA much harder to succeed, since the attack is meant to exploit overfitted training samples [50], [53], [65]. To understand the privacy risk in transfer learning, we need to ask whether it is possible to build a pre-trained model that appears to be well-generalized for the downstream task but stealthily renders the fine-tuning process vulnerable to inference attacks.

Our solution. Such a privacy risk turns out to be completely realistic, as demonstrated by our study, which utilizes *adversarial training* (AT) [40] on the pre-trained model to transfer the model’s weakness to the downstream without undermining the model’s accuracy. More specifically, AT has been extensively used for improving a model’s robustness [40], but is also found to cause the model to be more robust on the training data than on the testing data [51].

- The first two authors contributed equally to this work.

This robustness gap, which we term *robustness overfitting*, has been identified as a cause of increased membership leaks [55]. An interesting observation made in our research is that robustness overfitting can actually be transferred from an upstream pre-trained model to its downstream fine-tuned model, causing fine-tuning instances to be more robust than test instances, though the robustness gap shrinks (see Section 3). This allows the adversary to perform AT on a pre-trained model and then utilize this model to induce robustness overfitting on its fine-tuned model, so as to exploit the robustness gap for inferring the presence of a given sample in the fine-tuning set. Note that a robustness-overfitting model can still keep its generality, achieving high accuracy on the testing set.

This simple approach, however, does not work in practice, due to the diminishing robustness gap in the fine-tuned model, particularly between the fine-tuning samples and testing instances. Our study shows that regular AT performed on a model only causes its downstream counterpart to become slightly more robust on the fine-tuning instances than on testing ones, rendering these instances less differentiable in an MIA (see Section 5.5). Addressing this problem requires maximizing the robustness gap of the pre-trained model, without affecting its generality. This has been achieved in our research with a novel technique called *active robustness overfitting* (ARO), based upon the observation that the robustness gap can be maximized by AT when the model is overfitted to the training data (Section 3.2). ARO is designed to simulate the overfitting by adding the perturbations to model weights that minimize the losses of training instances, for the purpose of inducing an AT process to enlarge the robustness gap. The noise is later removed to preserve the model's performance on testing data. By iteratively applying this process to a model, we found that the robustness gap can be substantially extended. In our research, we further augment ARO with training data poisoning, a technique known to induce overfitting [60], which is found to also induce robustness overfitting. This is used to further increase the robustness gap with only a minor performance loss.

Notably, although prior work has identified that robustness overfitting would amplify membership leakage [55], it has focused on natural robustness overfitting caused by regular adversarial training algorithms, without recognizing actively induced robustness overfitting. In this paper, we reveal that while regular adversarial training algorithms do not pose a significant threat in transfer learning scenarios due to a low transferability rate, a carefully designed training algorithm can actively induce higher robustness overfitting and cause serious privacy leakage in these scenarios. While previous studies have also attempted to amplify membership leakage by manipulating pre-trained models, they often rely on prior knowledge of specific downstream target datasets [16], [37], [62]. In contrast, ARO introduces a versatile approach that is agnostic to particular target records or datasets.

Results. We implement ARO and evaluate its performance using models that were pre-trained on the Tiny ImageNet

and CIFAR-100 datasets. These models were subsequently fine-tuned for classification tasks on the CelebA, STL-10, and CINIC-10 datasets. In these experiments, we compromise the pre-trained models using ARO and assess privacy leakage from the models that were fine-tuned based on it. Our experimental results consistently demonstrate stable improvements with ARO in both the true-positive rate (TPR) at low false-positive rates (FPR) and the AUC (Area Under the Curve) score for MIA. For instance, in a black-box attack scenario where the downstream classifier is built on STL-10, ARO elevates its AUC score to 90.71%. In comparison, the AUC score is only 74.95% when employing models pre-trained in the normal manner, marking an improvement of over 15%, which is significant compared to the improvements in MIA reported by recent studies (under 10%) [20], [22]. Moreover, in a white-box attack scenario, at an FPR of 1%, the attack's TPR grows from 1.88% to 20.74%, an improvement by a factor of 11 $\times$. Meanwhile, ARO also improves the model's test accuracy on STL-10 by 13.4%. Such notable enhancements in performance could serve as a compelling reason for potential victims to use such a compromised pre-trained model.

Contributions. Our key contributions are outlined below:

- *New attack surface.* We develop ARO, an enhanced membership inference attack that takes place during the pre-training stage but targets the downstream task. Our study showcases a new yet realistic privacy threat to transfer learning, highlighting the importance of further understanding and mitigating the risk of this important learning paradigm.
- *Extensive empirical studies.* We perform comprehensive empirical studies of ARO on various datasets (including large datasets such as Tiny ImageNet and CINIC-10) and under diverse MIA scenarios. Our results demonstrate that our proposed method effectively amplifies the membership leakage while preserving satisfactory levels of test accuracy for the fine-tuned model on the downstream task.

2. Background

2.1. Transfer Learning

Transfer learning is a method commonly used in the machine learning landscape to alleviate the data and resource-intensiveness of training models from scratch, particularly in deep learning contexts [29], [46], [57]. Essentially, transfer learning involves leveraging a *pre-trained model*—a model that has been previously trained on a large, diverse dataset (referred to as the *pre-training dataset*)—to enhance new tasks that share certain similarities with the original task. This strategy allows organizations without massive datasets or computational resources to effectively construct accurate *fine-tuned models*—models derived from the pre-trained model and fine-tuned on a task-specific dataset (the *downstream dataset*)—for their unique application context.

Transfer learning can be formally defined as follows: Given a source domain and learning task, as well as a target domain and learning task, the goal of transfer learning is

to improve the learning of the target predictive function in the target domain. This is achieved by utilizing knowledge from the pre-trained model, originally trained on the source domain and its learning task. This method applies when the source and target domains are different, or the two learning tasks are distinct. If the new fine-tuned model has a similar domain or learning task as the pre-trained model, it can be constructed by fine-tuning its parameters to better align with the intended target task. This practice of using pre-trained models for downstream tasks forms the basis of transfer learning, enabling efficient and robust adaptation of deep learning models across a wide array of applications.

2.2. Membership Inference Attack

Membership inference attacks (MIAs) [50], [53], [65], are a form of privacy attack on machine learning models, where an adversary, with varying levels of access and knowledge about a target model f_{target} , seeks to ascertain whether a specific data instance x was part of the model's training set $\mathcal{D}_{\text{target}}^{\text{train}}$. Formally, an adversary constructs an attack model $f_{\text{attack}} : \mathcal{R} \rightarrow 0, 1$, leveraging any available resources, such as model responses to queries, shadow models, partial or complete knowledge about the target model's architecture, or a subset of the training data. In this context, $\mathcal{R}$ denotes the space of all possible responses from the target model, which can include model outputs, parameters, or any relevant metadata. The attack model's output for a given instance x determines the inferred membership status of x in the training set, where an output of 1 indicates that x is inferred to be a member of $\mathcal{D}_{\text{target}}^{\text{train}}$ (i.e., it was used in the training of f_{target}), and 0 indicates that x is inferred to be a non-member (i.e., it was not used in training f_{target}).

A successful MIA has significant privacy implications as it allows adversaries to gain access to sensitive information about the training data. In practical scenarios, x could represent a clinical record or an individual's information. Through MIA, attackers can determine whether this specific record or individual was part of a model's training dataset, potentially linked to a particular condition or disease. Such inference clearly violates privacy and confidentiality rules, emphasizing the necessity for robust measures against MIAs in privacy-sensitive machine learning applications.

2.3. Adversarial Training

Adversarial training (AT) is widely regarded as the most effective defense method against adversarial attacks [3], [4], [18], [45], [59]. AT essentially employs a "fight fire with fire" approach. It introduces adversarial examples into the training set, which are slight, carefully calculated modifications of the original inputs. These modifications are designed to mislead the model into making erroneous predictions. By learning from these adversarial examples, the model improves its ability to correctly classify such inputs in the future, thereby enhancing its overall resilience and robustness against adversarial attacks.

Formally, let $\mathcal{X}$ and $\mathcal{Y}$ be the input and output domains. Denote $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ as the training dataset with n samples, and $\mathcal{B}_\epsilon^p(x_i) = \{x'_i \in \mathcal{X} : \|x'_i - x_i\|_p \leq \epsilon\}$ as their adversarial regions, where ϵ is the maximum perturbation constraint. In AT, the training data are sampled entirely from these adversarial regions. Madry et al. [40] formulate AT as a min-max optimization problem:

$$\min_w \rho(w), \text{ where } \rho(w) = \sum_i \max_{x'_i \in \mathcal{B}_\epsilon^p(x_i)} \ell(f_w(x'_i), y_i), \quad (1)$$

where $\rho(w)$ is the *adversarial loss*, f_w represents the DNN classifier with weight w , and $\ell(\cdot)$ is the loss function. The inner maximization seeks an adversarial example x'_i that maximizes the loss, while the outer minimization optimizes the network parameters w to minimize the loss on these adversarial examples. To solve the inner maximization problem, one commonly used technique is *projected gradient descent* (PGD) [40]. It perturbs the normal example x_i for multiple steps K with a step size of α :

$$x_i^k = \Pi_{\mathcal{B}_\epsilon^p(x_i)}(x_i^{k-1} + \alpha \cdot \text{sign}(\nabla_{x_i^{k-1}} \ell(f_w(x_i^{k-1}), y_i))), \quad (2)$$

where x_i^k represents the adversarial example at step k , and $\Pi_{\mathcal{B}_\epsilon^p(x_i)}$ is the projection operator.

2.4. Robustness Overfitting

Similar to the natural training, overfitting is also a prevalent issue in the adversarial robust training of DNNs [48]. Beyond a certain point in AT, further training leads to a decline in robust accuracy on test data. This phenomenon, known as *robustness overfitting*, commonly occurs in DNN training, particularly with over-parameterized networks. This issue in AT negatively affects robustness generalization [9], [48]. Consequently, techniques to alleviate robustness overfitting have become essential for adversarial training.

Adversarial Weight Perturbation (AWP) [63] is an effective method to combat robustness overfitting by flattening the weight adversarial loss landscape, which is the landscape of the adversarial loss ($\rho(w)$) under varying degrees of perturbations to the model parameters, as illustrated by the two examples on the left in Section A, Figure 7. Robustness overfitting is mitigated through a dual-perturbation strategy that adversarially perturbs both inputs and weights:

$$\min_w \max_{v \in \mathcal{V}} \frac{1}{n} \sum_{i=1}^n \max_{\|x'_i - x_i\|_p \leq \epsilon} \ell(f_{w+v}(x'_i), y_i), \quad (3)$$

where n is the number of training examples, x'_i is the adversarial example of x_i , f_w is the DNN with weights w , $\ell(\cdot)$ is the loss function, ϵ is the maximum input perturbation constraint, and $\mathcal{V}$ is the feasible weight perturbation region, defined as $\{v \in \mathcal{V} : \|v\|_2 \leq \gamma \|w\|_2\}$ with γ being the weight perturbation size constraint. The inner maximization seeks adversarial examples x'_i within an ϵ -ball centered around the original examples x_i that maximize the classification loss ℓ . Conversely, the outer maximization seeks

weight perturbations v that maximize the loss ℓ on these adversarial examples to minimize the robust generalization gap. This paper utilizes a similar dual-perturbation approach that adversarially perturbs both inputs and weights, albeit in a reverse manner, to fulfill our attack objectives.

3. Threat Model and Key Observations

3.1. Threat Model

We focus on a transfer learning scenario in which a victim uses a pre-trained model to initialize their own downstream model, then fine-tunes it using their own training data. Intriguingly, we envision a situation where the provider of the pre-trained model has malicious intent, aiming to exploit this transfer learning scenario to infer the membership status of the downstream model's training data.

For example, a malicious agent might release a compromised model on popular public platforms such as HuggingFace¹, TensorFlow Model Garden², or Model Zoo³ for open access. A downstream user—like those utilizing services from Google Cloud or Microsoft Azure—may then download and fine-tune this compromised model using their own datasets. To increase the likelihood of adoption, the agent might offer a model tailored to specific domains or equipped with the latest architectural innovations.

Notably, pre-trained models are not exclusively developed by large companies. Platforms like HuggingFace host millions of models, many created by small companies or individual developers. Additionally, small companies or individuals often perform secondary development on pre-trained models originally released by major companies, such as Llama3, leading to new models that may have potential vulnerabilities. Victims often choose pre-trained models based on task-relevant keywords, which can inadvertently expose them to heightened risks of privacy leakage.

Attacker's goal. The attacker, in this case, the provider of the pre-trained model, has two primary objectives. The foremost goal is to subtly manipulate the pre-trained model so that any downstream model derived from it becomes more susceptible to membership inference attacks. This subtly is crucial, as the second objective of the attacker is to ensure that the manipulated pre-trained model does not degrade the performance of any downstream model. This is to ensure that the victim remains unaware of the manipulation and continues to use the pre-trained model in their applications.

Attacker's capabilities. The attacker is assumed to have full control over the training process of the pre-trained model, including hyperparameters, training data, and training methodology. However, we assume that the attacker is not allowed to change the model architecture, to keep the attack stealthy (changes to the model's code are much more likely to be detected). The attacker also does not

have control over the downstream model's training process, including the selection and handling of training data. This implies that the attack must be stealthy and not significantly impact the downstream model's performance to remain undetected. These assumptions create an attack vector that could potentially bypass existing defenses and leak sensitive information from the fine-tuned model's training data.

3.2. Key Observations

Song et al. [55] report that *robustness overfitting* leads to a higher success rate of MIAs. This inspires our research of inducing robustness overfitting into the pre-trained model to render its fine-tuned versions more vulnerable to MIAs. Our research revealed the privacy risks of the training data used for the fine-tuning of the model, while the manipulation of the pre-trained model is not easily detected because the introduced robustness overfitting does not weaken the performance of the pre-trained model or the fine-tuned model.

To achieve our goal, we attempt to first understand whether the robustness overfitting induced in a pre-trained model can be effectively transferred to its fine-tuned versions, and additionally, we explore the ways in which the robustness overfitting can be amplified. Through our research, we are able to validate these hypotheses in two key observations: 1) Robustness overfitting in a pre-trained model can be transferred to its fine-tuned counterparts. Specifically, when a pre-trained model exhibits robustness overfitting, its fine-tuned versions also demonstrate robustness overfitting to the fine-tuning data. Notably, even though the gap in robustness between the training and testing data of the fine-tuned model may be less than that of the pre-trained model, the phenomenon of overfitting persists. 2) The robustness overfitting exhibited by a pre-trained model can be amplified by implementing adversarial training on models that are already with *vanilla overfitting*. We elaborate on these two observations below.

Observation 1. We observe that if the robustness of a pre-trained model is considerably overfit (i.e., the gap between the robustness of the training data and the test data is large), the robustness of its fine-tuned version will also overfit: the robustness on the training data for fine-tuning is greater than that on any testing data, which implies that the robustness overfitting in a pre-trained model can be transferred to its derived models through fine-tuning.

To demonstrate this observation, we employ the models pre-trained on CIFAR-100 using the PGD adversarial training. The number of training epochs varies among {50, 60, 70, 80, 90, 100}, which show various levels of robustness overfitting into the model. Subsequently, we fine-tune these models on STL-10 for 100 epochs using a standard training procedure instead of AT, and then evaluate the robustness overfitting on the fine-tuned models. The experimental results indicate that there is a positive correlation between the levels of robustness overfitting (measured by the robustness gap between the training and testing data) in the pre-trained models and the fine-tuned models. A Pearson's

1. <https://huggingface.co>

2. <https://github.com/tensorflow/models>

3. <https://modelzoo.co>

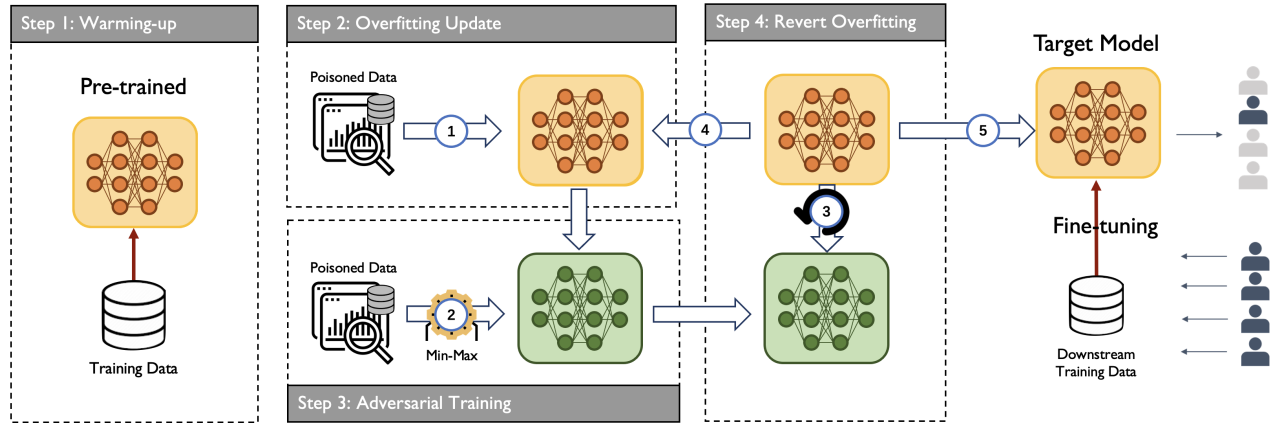


Figure 1: Overview of the ARO approach.

correlation analysis reveals a coefficient of 0.979, indicating a strong linear relationship between the two variables.

Notably, we emphasize the transferability of “robustness overfitting” rather than “robustness” itself. “Robustness” refers to a model’s improved performance on test data, whose transferability is well established. In contrast, “robustness overfitting” describes a scenario where a model appears more robust on its training data than on test data, creating a gap that can be exploited to distinguish training samples from test samples. While previous work has discussed the transferability of robustness (where a pre-trained robust model enhances the robustness of its fine-tuned version), no one has investigated the transferability of robustness overfitting, where the training-testing gap is passed from the pre-trained model to the fine-tuned model.

Robustness overfitting transfers from pre-trained models because the initial model weights already encode specific patterns and vulnerabilities that fail to generalize well under adversarial conditions. When a model is fine-tuned, the starting point provided by the pre-trained weights biases the optimization process. Essentially, if the pre-trained model has overfit to adversarial examples, these same brittle patterns persist during fine-tuning, especially when the fine-tuning dataset is limited. Consequently, the model carries forward its original robustness shortcomings into the new task, making it prone to the same generalization issues observed in the pre-trained model.

Observation 2. We observe that AT induces higher levels of robustness overfitting on a vanilla-overfitted model (i.e., its accuracy on training data is considerably higher than that on test data) than on a well-generalized model. The number of training epochs varies among {100, 120, 140, 160, 180, 200, 220}, which show various levels of vanilla overfitting. We then apply AT to these models using STL-10 for additional 100 epochs, and subsequently evaluate the levels of robustness overfitting of the resulting models. The experimental results indicate that there is a positive correlation between the levels of *vanilla overfitting* (measured by the gap of accuracy in the training data and

the testing data) on the models before AT and the levels of robustness overfitting on the models after AT. The Pearson’s correlation coefficient stands at 0.983, signaling a potent linear relationship between these two variables.

4. The ARO Attack

4.1. Intuition

Our general idea is to enhance robustness overfitting in the pre-trained model so that it can be transferred to the fine-tuned version, as noted in **Observation 1**, thereby making the training data used for fine-tuning more vulnerable to MIAs. The main challenge, however, is that the extent of robustness overfitting achieved solely through AT does not meet our needs (Orange bins in [Figure 2](#)). To address this, we intentionally induce vanilla overfitting into the pre-trained model before starting AT. This approach is guided by our **Observation 2**, which suggests that inducing vanilla overfitting can enhance robustness overfitting when the model undergoes AT. However, this method initially results in vanilla overfitting in the final model, potentially diminishing its performance due to the overfitting. To address this issue, we have developed an *active robustness overfitting* (ARO) technique. ARO leverages vanilla overfitting during the pretraining process to enhance robustness overfitting after AT, while ensuring that the final model does not suffer from vanilla overfitting, thus preventing a decline in performance due to overfitting. The design of ARO is inspired by the findings from a previous study, AWP [63], as introduced in [Section 2.4](#). A simplified illustration showing how AWP can mitigate robustness overfitting is provided in [Section A](#). ARO reconfigures the strategy of AWP for a different purpose, to effectively address this issue.

The process begins by adding perturbations to the weights of the pre-trained model to overfit it to the training data, followed by applying perturbations to inputs during AT. According to **Observation 2**, this sequence enhances robustness overfitting beyond what is achieved by direct

AT. It is important to note that in this step, unlike AWP, the perturbation introduced to the pre-trained model prior to AT is of an opposite nature to ours. This distinction arises because AWP is designed to achieve an opposite outcome—it aims to mitigate robustness overfitting, rather than enhance it. Upon completing this process, we employ a strategy similar to AWP; we remove the perturbations from the weights. This action eliminates the vanilla overfitting but retains the robustness overfitting. By repeatedly applying this process, we can induce a strong yet controllable level of robustness overfitting in the pre-trained model.

To further enlarge the robustness gap, our approach also allows a trade-off between model performance and the level of robustness overfitting, through data poisoning. Prior studies show that poisoning can raise a model's vanilla overfitting level [10], [60]. We observe that it also has the capability to enhance robustness overfitting, which can be utilized to make the fine-tuned model more vulnerable to MIA, at the cost of a minor performance loss.

4.2. Active Robustness Overfitting

Figure 1 illustrates the overall workflow of ARO. To efficiently achieve the robustness overfitting of pre-trained models, ARO employs a double perturbation approach, iterating over four steps of model updating as presented below.

Step 1: Warming-up. The pre-trained model first undergoes a warming-up phase with standard training to allow the accuracy (ACC) to reach the desired standard. We denote the model after warm-up as f_w .

Step 2: Best-case model perturbation for vanilla overfitting. We select a batch of training data X_{batch} , on which we compute the gradients using the *stochastic gradient descent* (SGD) algorithm. Subsequently, we perturb the model weights by using a larger step size along the gradients' direction, aiming at overfitting the model to the selected batch of training data. We denote the perturbation as δ , and the model after perturbation as $f_{w+\delta}$.

Step 3: Adversarial training (AT). To accomplish robustness overfitting to the training data, we perform the adversarial training on X_{batch} using the vanilla-overfitted model $f_{w+\delta}$. We denote the model after AT as $f_{w+\delta+\beta}$, where the β is the weight update yielded from AT.

Step 4: Revert perturbation. In Step 2, we perform a model perturbation to enable the robustness overfitting in Step 3. However, we do not intend for the pre-training model to be vanilla overfitted. Therefore, in this step, we revert the update of model weight in Step 2: the final model is $f_{w+\delta+\beta-\delta} = f_{w+\beta}$.

Remarks. The above workflow operates iteratively: after Step 4 is completed, the process goes back to Step 1 while a different batch of training data is randomly selected in each iteration. The total number of iterations is given as the input to the algorithm so that the training is completed after the specific number of iterations. Notably, our approach is different from the conventional technique used

for improving the model's robustness generalizability, such as AWP [63], which intensifies the worst-case perturbations by performing the adversarial training before reverting the perturbations. By improving the adversarial robustness on the selected batch of training data through the best-case model perturbation (i.e., in an attempt to render the vanilla overfitting into the training data) and AT, ARO actively induces robustness overfitting and thus limits the model's robustness generalizability. Below, we present specific optimization techniques used in Step 2-4.

Best-case model perturbation. The *best-case model perturbation* δ is defined as:

$$\arg \max_{\delta} \left[\sum_{x_i, y_i \in \mathcal{D}} \ell(f_w(x_i), y_i) - \ell(f_{w+\delta}(x_i), y_i) \right] \quad (4)$$

$$\text{s.t. } \|\delta\|_2 = \gamma$$

where ℓ is the loss function, (e.g., the MSE for regression tasks or CE for classification tasks). $\mathcal{D}$ is the training dataset. Intuitively, the overfitting perturbation (of a constant size γ) results in the lowest loss $\ell(w + \delta)$ when the weights are perturbed by δ .

In practice, during the optimization, we employ a batch scheme to accommodate optimizers such as SGD. Consequently, in each iteration, $\ell(f_w)$ and $\ell(f_{w+\delta})$ in Equation 4 represent the loss measured over a batch of training data:

$$\arg \max_{\delta} \left[\sum_{i \in \text{batch}} \ell(f_w(x_i), y_i) - \ell(f_{w+\delta}(x_i), y_i) \right] \quad (5)$$

$$\text{s.t. } \|\delta\| = \gamma$$

Adversarial training. After perturbing the model, we perform AT on the model $f_{w+\delta}$ using the PGD [40] optimization:

$$w^* = \arg \min_{w+\delta} \frac{1}{n} \sum_{i=1}^n \max_{\|x'_i - x_i\|_p \leq \epsilon} \ell(f_{w+\delta}(x'_i), y_i) \quad (6)$$

Revert perturbation. After completing the adversarial training, it becomes necessary to revert the best-case perturbation from the weights w^* .

$$w_{new} = w^* - \delta \quad (7)$$

Overall Algorithm. The algorithm for the proposed ARO attack can be found in Algorithm 1, as detailed in Appendix Section D.

4.3. Membership Inference Strategy Selection

Song et al. [55] found that input robust models are more susceptible to privacy attacks. They introduced a new inference strategy utilizing the model's predictions on adversarial examples, denoted as $\mathcal{I}_A$. In contrast, the basic inference strategy using clean inputs is represented as $\mathcal{I}_B$. Their findings indicate that $\mathcal{I}_A$ outperforms $\mathcal{I}_B$ when applied to robust models, whereas $\mathcal{I}_B$ is more effective than $\mathcal{I}_A$ with naturally trained (undefended) models. In our case, despite applying adversarial training to the upstream

model to induce robustness overfitting, the model fine-tuned with downstream data tends to be less robust, especially compared to a downstream model directly enhanced with AT. Therefore, we opt for $\mathcal{I}_B$ as our inference strategy. A comprehensive comparison between $\mathcal{I}_A$ and $\mathcal{I}_B$ can be found in Section C.

5. Evaluation

5.1. Experimental Setup

Datasets. We use the following five datasets for evaluation.

- **CIFAR-100** [30]: This dataset contains 50,000 training images and 10,000 testing images. Each image has a size of $32 \times 32 \times 3$ and belongs to one of 100 classes.
- **Tiny ImageNet** [33]: This dataset contains 100,000 training images and 10,000 testing images. Each image has a size of $64 \times 64 \times 3$ and belongs to one of 200 classes.
- **CelebA** [38]: This dataset contains 202,599 face images annotated by 40 attributes. In our experiment, we train a eight-class classifier to predict the top three most balanced attributes including Heavy Makeup, Mouth Slightly Open, and Smiling.
- **STL-10** [11]: This dataset contains 13,000 color images. Each image has a size of $96 \times 96 \times 3$ and belongs to one of 10 classes, and each class contains 1,300 images.
- **CINIC-10** [12]: This dataset is compiled from images selected and downsampled from the ImageNet database [14]. This dataset includes 270,000 color images.

CIFAR-100 and Tiny ImageNet serve as the upstream datasets, while CelebA, STL-10, and CINIC-10 are used as the downstream datasets. Each original downstream dataset, denoted by $\mathcal{D}$, is partitioned into four equally sized, disjoint subsets: $\mathcal{D}_{\text{target}}^{\text{train}}$, $\mathcal{D}_{\text{target}}^{\text{test}}$, $\mathcal{D}_{\text{shadow}}^{\text{train}}$, and $\mathcal{D}_{\text{shadow}}^{\text{test}}$. The subsets $\mathcal{D}_{\text{target}}^{\text{train}}$ and $\mathcal{D}_{\text{target}}^{\text{test}}$ serve as the training and testing sets for the target model, respectively, while $\mathcal{D}_{\text{shadow}}^{\text{train}}$ and $\mathcal{D}_{\text{shadow}}^{\text{test}}$ are used as the training and testing sets for the shadow model.

Pre-training procedure. We use CIFAR-100 and Tiny ImageNet as our pre-training datasets and employ ARO to train a PreAct-ResNet18 [19] model as the pre-trained model. We compute the mean and standard deviation on the training set to normalize the input images. We adopt the cross-entropy loss function and the SGD optimizer with momentum with a step-wise learning rate decay. Specifically, the model is trained for 200 epochs with the learning rate initialized at 0.01 and divided by 10 after 100 and 150 epochs. The batch size is set to 128, and the weight decay is set to 1×10^{-4} .

Fine-tuning procedure. Given a pre-trained model, we use it to train classifiers for the downstream datasets – CelebA, STL-10, and CINIC-10. We utilize the pre-trained model’s parameters as the initial parameters of the downstream classifier and then train the classifier on the downstream dataset. We adopt the cross-entropy loss function and the SGD optimizer with momentum when training a downstream classifier. The model is fine-tuned for 100 epochs with the learning rate initialized at 0.01 and divided by 10 after 50

and 75 epochs. The momentum is set to 0.9, and the weight decay is set to 5×10^{-4} for all experiments.

MIA configuration and evaluation metrics. According to the different knowledge levels the attacker has about the downstream model, we consider four types of MIAs:

- **Black-Box/Shadow** [50]: In this scenario, the attacker only has black-box access to the target model with a local shadow dataset and lacks knowledge of the training dataset of the target model. Note that the attacker can obtain confidence scores returned by the target model.
- **Black-Box/Partial** [50]: The attacker has black-box access to the target model and a partial training dataset of the target model. In this scenario, the adversary does not need to train a shadow model; rather, they use the partial training dataset as the ground truth for membership and directly train their attack model.
- **White-Box/Shadow** [43]: The attacker has white-box access to the target model but only has a local shadow dataset and lacks knowledge of the training dataset of the target model. As the adversary has white-box access to the target model, they can also exploit the target sample’s gradients with respect to the model parameters, embeddings from different intermediate layers, classification loss, and prediction posteriors.
- **White-Box/Partial** [43]: The attacker has white-box access to the target model and has a partial training dataset of the target model. The attack methodology here is almost identical to the black-box counterpart.

We then establish two types of attack models: one for the black-box and the other for the white-box setting. For black-box, our attack model has two inputs: the target sample’s ranked posteriors and a binary indicator on whether the target sample being predicted correctly. Each input is first fed into a different 2-layer MLP (Multilayer Perceptron), then the two obtained embeddings are concatenated together and fed into a 4-layer MLP. For white-box, we have four inputs for this attack model, like the one used by Nasr et al. [43], including the target sample’s ranked posteriors, classification loss, gradients of the parameters of the target model’s last layer, and one-hot encoding of its true label. Each input is fed into a different neural network, and the resulting embeddings are concatenated together as input to a 4-layer MLP. We use ReLU as the activation function for the attack models. The batch size is set to 64, and cross-entropy is the loss function. We use Adam as the optimizer, with learning rate of $1e-5$. The attack model is trained for 50 epochs. We adopt downstream accuracy and MIA’s AUC (area under the ROC curve) score as the evaluation metrics.

Attack variants. We consider two variants of the proposed ARO attack to evaluate its effectiveness across different levels of aggression and stealthiness: a more stealthy version, referred to as ARO_α , and a more aggressive version, referred to as ARO_β . For ARO_α , we take L_2 threat model for adversarial training [40] with $\epsilon = 128/255$, step size $\alpha = 15/255$, and PGD steps $K = 1$. For overfitting perturbation magnitude, we adopt $\gamma = 1 \times 10^{-3}$ by default. Conversely, ARO_β employs the same parameters as ARO_α ,

TABLE 1: Membership inference (MI) results, downstream accuracy, vanilla overfitting rate, and robustness overfitting rate across three different downstream datasets under various upstream training methods.

Downstream Dataset	Upstream Training Method	Vanilla Overfitting	Robustness Overfitting	Black-Box\Shadow		Black-Box\Partial		White-Box\Shadow		White-Box\Partial	
				TPR	AUC	TPR	AUC	TPR	AUC	TPR	AUC
CelebA	Normal Pre-training	0.2269	0.1203	0.0087	0.7037	0.0100	0.7149	0.0467	0.7614	0.0547	0.7663
	Poisoning Only	0.2274	0.1284	0.0113	0.7237	0.0140	0.7254	0.0464	0.7853	0.0580	0.7954
	ARO $_{\alpha}$ (ours)	0.2384	0.1354	0.0139	0.7167	0.0153	0.7341	0.0478	0.7863	0.0639	0.7972
	ARO $_{\beta}$ (ours)	0.2365	0.1586	0.0180	0.7404	0.0188	0.7429	0.0656	0.7926	0.0733	0.8034
STL-10	Normal Pre-training	0.1550	0.0563	0.0065	0.7495	0.0089	0.7538	0.0092	0.8018	0.0188	0.8449
	Poisoning Only	0.2553	0.0944	0.0065	0.8064	0.0123	0.8158	0.0148	0.8298	0.0585	0.8920
	ARO $_{\alpha}$ (ours)	0.2449	0.1286	0.0175	0.8335	0.0185	0.8414	0.0199	0.8653	0.0338	0.8655
	ARO $_{\beta}$ (ours)	0.2918	0.1303	0.0283	0.9071	0.0360	0.9151	0.0702	0.9412	0.2074	0.9501
CINIC-10	Normal Pre-training	0.2335	0.1170	0.0000	0.7865	0.0176	0.8045	0.0596	0.8221	0.0635	0.8339
	Poisoning Only	0.2998	0.1173	0.0000	0.8226	0.0243	0.8232	0.0837	0.8551	0.0862	0.8656
	ARO $_{\alpha}$ (ours)	0.2594	0.1390	0.0000	0.8175	0.0246	0.8396	0.0838	0.8547	0.1116	0.8680
	ARO $_{\beta}$ (ours)	0.2935	0.1597	0.0011	0.8662	0.0182	0.8876	0.0998	0.9025	0.1147	0.9129

but incorporates data poisoning. Specifically, we poison half of the dataset by flipping all labels to a randomly chosen class, achieving a poisoning ratio of 0.5. We adopt random labeling in our data poisoning approach, which we found to be the most effective as detailed in Section 5.7.

We observe that the training time for both ARO $_{\alpha}$ and ARO $_{\beta}$ is 1.6 times that of normal pre-training. This increase is typically manageable as it represents a one-time cost.

5.2. ARO Substantially Amplifies Leakage

In this section, we empirically validate the overall effectiveness of ARO and conduct ablation studies to illustrate the effectiveness of each component.

Setup. We conduct experiments on four different MIA scenarios across three distinct downstream tasks (CelebA, STL-10, and CINIC-10) under four unique pre-training settings: (i) Normal Pre-training: pre-training on the clean dataset utilizing standard training procedures; (ii) Poisoning Only: pre-training on the poisoned dataset with a poisoning ratio of 0.5; (iii) ARO $_{\alpha}$: pre-training with both input perturbation and weight perturbation on the clean dataset; (iv) ARO $_{\beta}$: pre-training with both input perturbation and weight perturbation on the poisoned dataset. In this experiment, we use CIFAR-100 as the fixed dataset for pre-training.

Observations. Table 1 illustrates the attack AUC for different settings. In general, we observe that ARO significantly increases the AUC score of the MIA attacks in all settings. Using the most common attack setting, *Black-Box/Shadow*, as an example, when the downstream task is STL-10, ARO $_{\beta}$ boosts the AUC to 90.71%. By contrast, the AUC is only 74.95% when employing models pre-trained normally, marking an improvement of over 15%. Regarding worst-case leakage, ARO $_{\beta}$ boosts the TPR@1%FPR from 0.65% to 2.83%, an improvement by a factor of over 4. When the downstream task is CINIC-10, ARO $_{\beta}$ elevates the AUC from 78.65% to 86.62%, an improvement of 8%.

TABLE 2: ARO preserves downstream accuracy.

Upstream Training Method	Downstream Dataset		
	CelebA	STL-10	CINIC-10
From Scratch	0.7551	0.5997	0.6699
Normal Pre-training	0.7730	0.8449	0.7661
Poisoning Only	0.7726	0.7446	0.6999
ARO $_{\alpha}$ (ours)	0.7615	0.7551	0.7403
ARO $_{\beta}$ (ours)	0.7635	0.7337	0.7062

Meanwhile, ARO $_{\beta}$ boosts the TPR@1%FPR from 0% to 0.11%. Although this increase is modest, it could be disastrous if real sensitive data falls within this 0.11%, as this portion of the data would be nearly fully exposed. Regarding the most powerful threat model, *White-Box/Partial*, when the downstream task is STL-10, ARO $_{\beta}$ boosts the AUC from 84.49% to 95.01%, an improvement of over 10%. Additionally, the TPR@1%FPR is boosted from 1.88% to 20.74%, an improvement by a factor of over 10. This demonstrates that ARO $_{\beta}$ can be highly effective, even when the privacy leakage is already substantial.

We note that ARO $_{\alpha}$ primarily enhances effectiveness by increasing robustness overfitting, whereas *Poisoning Only* chiefly boosts effectiveness by augmenting vanilla overfitting with a lesser impact on robustness overfitting. This difference may be why ARO $_{\alpha}$ improves the effectiveness-stealthiness tradeoff, unlike the poisoning approach. Furthermore, we observe that integrating the poisoning approach with ARO $_{\alpha}$ is a necessary strategy, as their combination (ARO $_{\beta}$) leads to a notable improvement over using each separately. Additionally, ARO $_{\beta}$ consistently achieves the highest robustness overfitting, suggesting that data poisoning can further help amplify the robustness overfitting.

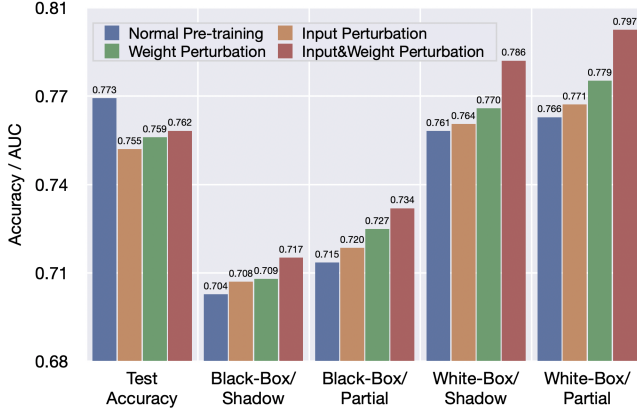


Figure 2: The impact of perturbation type.

5.3. ARO Preserves Downstream Accuracy.

Table 2 compares the accuracy of downstream classifiers trained upon pre-trained models, which were trained using four distinct upstream methods mentioned previously, with a classifier trained from scratch (using randomly initialized weights, i.e., no pre-training). This experiment involves all the three downstream datasets. Generally, we observe that ARO consistently enhances the accuracy of the downstream classifier compared to training from scratch across all settings. For instance, in the case of STL-10, ARO_{β} raises the test accuracy from 59.97% to 73.37%, marking an improvement of 13.4%. When the downstream task is CINIC-10, ARO_{β} boosts the accuracy from 66.99% to 70.62%, an improvement of 3.63%. Identifying the optimal accuracy achievable with pre-training can be challenging for users. The notable accuracy gains with ARO_{β} highlight its advantage over training from scratch, potentially encouraging users to adopt the attacker’s pre-trained model due to these performance benefits.

We also notice that, compared to a normal pre-trained model, ARO_{β} shows a slight decrease in downstream accuracy. For example, with CINIC-10 as the downstream dataset, ARO_{β} achieves a downstream test accuracy of 70.62%, whereas a normal pre-trained model reaches 76.61%. However, we argue that this minor reduction is acceptable, as the victim may not be aware of the accuracy level achievable with a normal pre-trained model. If using a compromised pre-trained model results in higher test accuracy compared to training from scratch, victims could be inclined to use the pre-trained model, as this significant performance improvement provides a compelling reason for its adoption. Moreover, we observe that ARO_{α} yields higher downstream accuracy while maintaining similar attack effectiveness as the *Poisoning Only* approach (refer to Table 1). This indicates that the improvement in the effectiveness-stealthiness tradeoff is primarily attributed to ARO_{α} .

5.4. Impact of Upstream Dataset

Setup. In this experiment, we explore the impact of public dataset selection on ARO. We evaluate two public datasets: Tiny ImageNet and CIFAR-100.

Observations. The results, as illustrated in Table 3, demonstrate the influence of public dataset selection on the effectiveness of ARO. Overall, ARO proves effective across both configurations, highlighting its adaptability and stability. Notably, pre-training with Tiny ImageNet appears to be more appropriate for attacking CINIC-10, while pre-training with CIFAR-100 shows better suitability for attacking CelebA and STL-10. Take CINIC-10 as an example: in the *Black-Box/Shadow* scenario, pre-training with Tiny ImageNet achieves a TPR@1%FPR of 0.44%, whereas pre-training with CIFAR-100 achieves only 0.11%; in the *White-Box/Partial* scenario, pre-training with Tiny ImageNet achieves a TPR@1%FPR of 13.40%, while pre-training with CIFAR-100 achieves 11.47%. However, in general, we observe that the effectiveness of ARO does not vary significantly across different upstream datasets, and no single dataset consistently outperforms the others.

As a result, in the subsequent sections, we use CIFAR-100 as the fixed dataset for pre-training to deepen our understanding of our approach.

5.5. Contributions of Perturbation Types

Recall that the proposed ARO method utilizes both input and weight/overfitting perturbations. In this section, we aim to understand the contribution of each type of perturbation.

Observations. Figure 2 illustrates the influence of perturbation type on effectiveness and stealthiness. In this experiment, we fix the downstream dataset as CelebA and maintain the upstream dataset as clean (without poisoning). In general, the experimental results suggest that both input and weight perturbations can amplify the membership leakage with minimal impact on downstream accuracy. When we apply both types of perturbations simultaneously, they mutually benefit and yield a substantial amplification in membership leakage. Furthermore, combining these two types of perturbations does not lead to a further decrease in downstream accuracy. As a result, this combination indeed improves the effectiveness-stealthiness tradeoff.

In addition, we observe that introducing weight/overfitting perturbation consistently enhances effectiveness far more than adding input perturbation alone does. Adding input perturbation alone, also known as adversarial training (AT), usually yields limited improvement in effectiveness. As a result, we posit that a standard robust encoder as a service [8], [26], [28], [47] raises minimal privacy concerns. In other words, transferable input robustness does not inherently conflict with privacy concerns in downstream tasks. Only when the robust encoder is maliciously trained (for instance, using the proposed ARO method) to achieve abnormal robustness overfitting should the privacy concerns of the downstream task be escalated. As a result, future

TABLE 3: The impact of upstream dataset.

Downstream Dataset	Upstream Dataset	Upstream Training Method	Black-Box\Shadow		Black-Box\Partial		White-Box\Shadow		White-Box\Partial	
			TPR	AUC	TPR	AUC	TPR	AUC	TPR	AUC
CelebA	CIFAR-100	Normal Pre-training	0.0087	0.7037	0.0100	0.7149	0.0467	0.7614	0.0547	0.7663
		ARO _{β} (ours)	0.0180	0.7404	0.0188	0.7429	0.0656	0.7926	0.0733	0.8034
	Tiny ImageNet	Normal Pre-training	0.0108	0.6997	0.0126	0.7078	0.0443	0.7672	0.0551	0.7752
		ARO _{β} (ours)	0.0112	0.7144	0.0146	0.7237	0.0513	0.7908	0.0556	0.8142
STL-10	CIFAR-100	Normal Pre-training	0.0065	0.7495	0.0089	0.7538	0.0092	0.8018	0.0188	0.8449
		ARO _{β} (ours)	0.0283	0.9071	0.0360	0.9151	0.0702	0.9412	0.2074	0.9501
	Tiny ImageNet	Normal Pre-training	0.0102	0.7134	0.0126	0.7250	0.0222	0.7638	0.0400	0.8250
		ARO _{β} (ours)	0.0139	0.8105	0.0154	0.8334	0.0322	0.8876	0.0646	0.9208
CINIC-10	CIFAR-100	Normal Pre-training	0.0000	0.7865	0.0176	0.8045	0.0596	0.8221	0.0635	0.8339
		ARO _{β} (ours)	0.0011	0.8662	0.0182	0.8876	0.0998	0.9025	0.1147	0.9129
	Tiny ImageNet	Normal Pre-training	0.0000	0.7814	0.0147	0.7866	0.0619	0.8153	0.0631	0.8374
		ARO _{β} (ours)	0.0044	0.8675	0.0260	0.8808	0.1329	0.8896	0.1340	0.9041

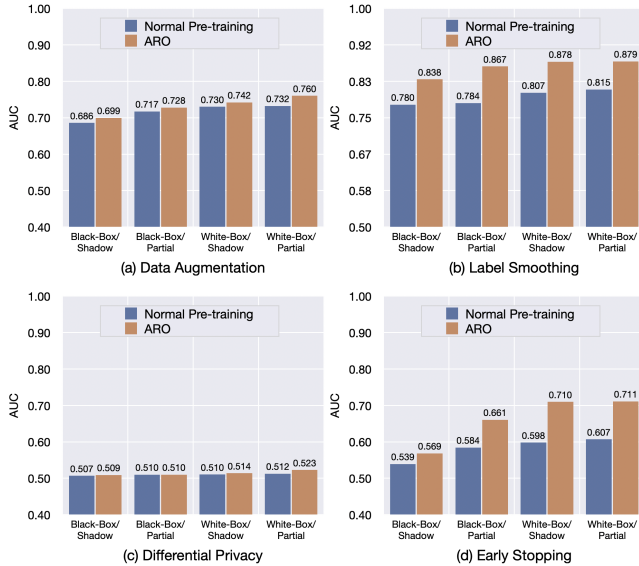


Figure 3: Effectiveness of the attack against four downstream defense strategies: data augmentation, label smoothing, differential privacy, and early stopping.

work could focus on detecting such abnormal robustness overfitting in pre-trained models to better safeguard the privacy of downstream tasks.

5.6. Robustness Against Existing Defenses

In this section, we conduct experiments to assess whether ARO can effectively counter existing defense strategies.

Existing Defenses. We have chosen four representative defense methods applicable in both black-box and white-box MIA settings: data augmentation [27], label smoothing [58], differential privacy [15] and early stopping [31].

- **Data Augmentation** [27] enhances the training dataset by slightly altering the original data samples, thereby helping to mitigate vanilla overfitting.

- **Label Smoothing** [58] serves as a regularization technique to moderate the target model’s overconfidence, implemented with a smoothing factor set to 1×10^{-3} .

- **Differential Privacy** [15] has been the golden standard for effective control of the risks of exposing training examples, and has already been adopted by various machine learning tasks [13], [34], [41], [68] for protection. Differentially private stochastic gradient descent (DP-SGD) [1] stands out as a prevalent privacy-preserving machine learning algorithm. In our experiment, we utilize the DP-SGD optimizer implemented using Opacus 1.4.0. We set $\delta = 1 \times 10^{-5}$ and allocate a privacy budget of $\epsilon = 300$, with the clipping threshold C fixed at 1.5.

- **Early Stopping** [31] is a widely used method to prevent overfitting. During training, for each epoch, we allocate 10% of the training data for validation and use the remaining 90% for training purposes. We track the validation loss, and if it fails to decrease over three consecutive epochs, we terminate the training process.

Observations. We evaluate the four defenses on CINIC-10 dataset and summarize the results in Figure 3. Comparing membership exposure between unprotected (Table 1) and protected models (Figure 3), all the four defenses can help weaken the membership exposure.

Furthermore, it is noticeable that ARO still results in higher membership leakage compared to a normal pre-trained model, even when defense strategies are implemented and overall membership leakage is reduced. This could be because current defense strategies typically concentrate on mitigating vanilla overfitting, while robustness overfitting remains largely unaddressed. As a result, ARO effectively raises the bar for MIA defense. With the use of ARO, a more robust defense is required to achieve a similar level of control over membership leakage as would

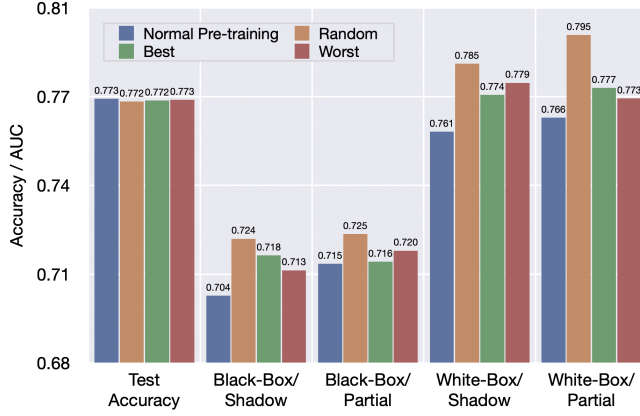


Figure 4: The impact of poisoning method.

be without ARO. Notably, differential privacy only provides an upper bound on worst-case privacy loss and does not directly restrict how much individual training data is memorized by the model. By promoting robustness overfitting, our approach encourages the model to capture more detailed, training-specific patterns, which increases its vulnerability to MIAs, even when DP noise is applied. However, when the DP noise is set at a high level, the overall leakage is reduced, limiting our approach’s impact, while also significantly degrading model utility.

5.7. Choices of Poisoning Methods

To understand how data poisoning contributes to ARO, we start with understanding the impact of poisoning method.

Candidates. We consider three poisoning methods:

- *Random*: mislabel the samples with incorrect labels at random (100 choices in CIFAR-100).
- *Best*: mislabel the samples with the most likely incorrect class for that sample (as predicted by a pre-trained model).
- *Worst*: mislabel the samples with the least likely class.

Setup. In this experiment, we fix the upstream dataset as CIFAR-100 and the downstream dataset as CelebA and the training process is kept clean (without any input or weight perturbations), with only data poisoning introduced. The poisoning ratio is set at 0.5, meaning that the dataset consists of 50k samples with 25k clean samples and 25k mislabeled (poisoned) samples.

Observations. Figure 4 illustrates the impact of the poisoning method on effectiveness and stealthiness. The experimental results suggest that these three poisoning methods provide similar stealthiness (downstream test accuracy), but the random poisoning approach achieves the highest effectiveness. This finding also reveals that the random poisoning approach provides the optimal effectiveness-stealthiness trade-off. Consequently, we adopt random poisoning as the default setting for our ARO attack.

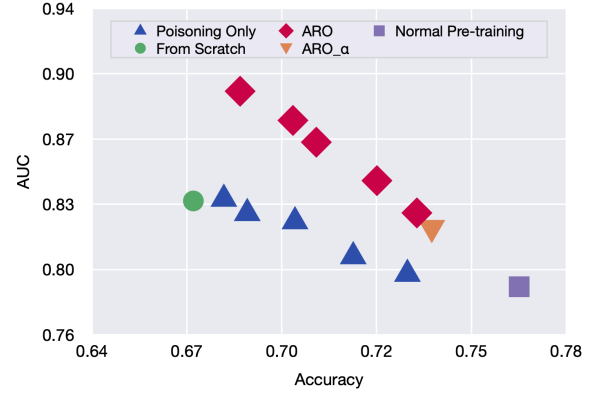


Figure 5: The impact of the poisoning ratio on both the Poisoning Only attack and the ARO attack. The results of the Poisoning Only attack are represented by blue triangles, while the results of the ARO attack are depicted by red diamonds. The impact of ARO_{α} , a special case of ARO, is shown by an orange inverted triangle. Results obtained using a normally pre-trained model are illustrated by a green circle, and those from training from scratch are symbolized by a purple rectangle.

5.8. Impact of Poisoning Ratio

Setup. In this experiment, we consider five poisoning ratios in the upstream: 0.1, 0.3, 0.5, 0.6, and 0.7. The downstream dataset is fixed at CINIC-10. The results of the *Poisoning Only* attack are denoted by blue triangles, and the results of the ARO attack are denoted by red diamonds. The result of using a normally pre-trained model is denoted as a green circle, and the result of training from scratch is marked as a purple rectangle.

Observations. Figure 5 demonstrates the impact of the poisoning ratio on both the *Poisoning Only* and the ARO attack. We observe that training from scratch aligns closely with the *Poisoning Only* attack at different poisoning ratios. This suggests that the poisoning approach typically sacrifices attack stealthiness (downstream test accuracy) for increased effectiveness and hence does not offer any advantages in the effectiveness-stealthiness trade-off. On the other hand, all the results of the ARO attack are in the top-right region compared to the *Poisoning Only* attack. As a result, the ARO attack can indeed improve the effectiveness-stealthiness trade-off, with the poisoning ratio acting as a controller to adjust this trade-off.

In conclusion, both the poisoning approach and the ARO_{α} approach have their limitations. The ARO_{α} approach has a clear upper limit and thus lacks scalability. On the other hand, the poisoning approach is highly scalable; it can always trade stealthiness for greater effectiveness without a clear limitation. However, the poisoning approach can only achieve a balanced trade with no net gain in the effectiveness-stealthiness trade-off, while ARO_{α} is an effective method to enhance the trade-off by utilizing robustness overfitting instead of vanilla overfitting. Therefore, both approaches are integral to the proposed ARO attack.

6. Potential Countermeasures

Prevention-based countermeasures. Recall that ARO primarily operates by actively promoting the transfer of robustness overfitting from the pre-trained model. We also find that existing defense techniques are incapable of counteracting the robustness overfitting transferred from the upstream model. This could be because these techniques lack an adversarial training process, thus they can only impact vanilla overfitting and not robustness overfitting. Consequently, a straightforward countermeasure could be to apply AT in the downstream model, which could use accumulated downstream robustness to mask the minor robustness effect transferred from the upstream model. Another avenue of defense is to utilize robustness overfitting countermeasures [9], [56], [61], [63], [67], [70]. We evaluated one such method – adversarial weight perturbation (AWP) [63] – which aims to reduce the robustness generalization gap by flattening the weight loss landscape.

We evaluate these two potential countermeasures on the CINIC-10 dataset and summarize the results in Figure 6. A comparison of membership exposure between unprotected models (Table 1) and those with implemented defenses (Figure 6) reveals that these defenses do help in reducing membership exposure. However, a noticeable gap persists between the pre-trained model trained with ARO and the normal pre-trained model. This suggests that the amplification effect on membership exposure caused by our attacks is not fully mitigated. Additionally, when downstream models apply AT or AWP for protection, the inference strategy $\mathcal{I}_B$ proposed by [55] could potentially increase attack effectiveness. We reserve this line of exploration for future work.

It is worth noting that these strategies typically involve adversarial training procedures, which are significantly more computationally demanding and time-consuming than regular training. As such, they may not be practical for all applications due to their increased computational requirements. Therefore, the development of more efficient defense strategies against such attacks continues to be a crucial area for future research.

Detection-based countermeasures. Another potential approach is to detect such malicious pre-trained models. We can consider two types of defender capabilities: (i) if the defender has access to part of the training and testing set of the pre-training model, they can directly check for abnormal robustness overfitting as an indicator of privacy risk; (ii) if the defender has no information about the upstream datasets, they can use a small downstream dataset to evaluate the level of transferable robustness overfitting as an indicator of potential privacy risk. The main challenge with these approaches lies in defining a threshold for what level of robustness overfitting should be considered as high risk. Setting this threshold would require a deep understanding of the typical patterns of robustness overfitting under different conditions, and would likely involve both empirical studies and theoretical analysis.

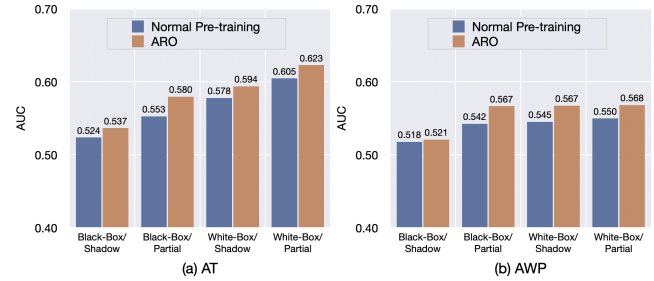


Figure 6: Attack effectiveness when AT and AWP are taken in downstream as defenses.

7. Related Work

Membership inference attacks. Shokri et al. [53] propose the first membership inference attack against black-box ML models: they train multiple shadow models to simulate the target model and use multiple attack models to conduct the inference. Salem et al. [50] later relax several key assumptions from [53]; namely, using multiple shadow models, the knowledge of the target model structure, and having a dataset from the same distribution as the target model's. Yeom et al. [65] assume that the adversary knows the target model's training dataset's distribution and size, and they collude with the training algorithm. Both [50] and [65] are close in performance to Shokri et al.'s attacks [53]. In this paper, we implement the attack proposed by Salem et al. [50], i.e., one shadow model, one attack model, and a shadow dataset.

Previous work [50], [53] suggests that vanilla overfitting is a major factor contributing to the success of MIAs. Consequently, some studies have leveraged strategies that can enhance vanilla overfitting to improve MIAs [10], [60]. However, to the best of our knowledge, no research has yet exploited robustness overfitting for more advanced attacks.

Furthermore, recent research [55] indicates that adversarial robust models are significantly more susceptible to membership inference attacks. However, these studies focus exclusively on passive robustness overfitting and do not investigate active robustness overfitting, which is the primary focus of our paper. Moreover, they do not consider the transfer learning scenario where robustness overfitting does not directly accumulate on the target training data. In this paper, we pioneer the study of the transferability of this threat, which has been largely under-explored in the past.

The most closely related works are those by Wen et al. [62], Liu et al. [37] and Feng et al. [16], which all aim to enhance privacy attacks in downstream tasks by poisoning the upstream model. However, these approaches rely on additional and less realistic assumptions. Wen et al. [62] and Feng et al. [16] assume that the adversary controls the pre-training process and has prior knowledge of specific target data points. Similarly, Liu et al. [37] assumes that the adversary controls the pre-training process and has

access to an auxiliary dataset from the same distribution as the downstream training data. In contrast, our attack is untargeted and requires no prior knowledge of the sensitive downstream training data, making it a more realistic and challenging scenario compared to these studies.

MIA defenses. Existing defenses against MIAs primarily aim to enhance the robustness of the target model, employing methods such as confidence score masking [25], [35], [53], regularization [27], [42], [53], or differential privacy [6], [23], [52]. We assessed the effectiveness of our proposed ARO attack against four representative defense methods: data augmentation [27], label smoothing [58], differential privacy [15], and early stopping [31]. The experimental results indicate that the proposed ARO effectively raises the bar for MIA defense.

Attacks and defenses in transfer learning. Some research has also focused on membership inference attacks in transfer learning scenarios. Liu et al. [36] demonstrated a membership inference attack against pre-trained encoders, which can infer whether an input was part of the encoder's training dataset. Concurrently, He et al. [21] presented a mitigation strategy that can counteract such threats. However, our work shifts the focus onto the privacy risks of the downstream tasks rather than the pre-trained encoders. We are the first to study the potential privacy risks from malicious pre-trained models to downstream tasks.

Another branch of research concentrates on malicious pre-trained encoders that can insert Trojans into downstream tasks. Jia et al. [24] proposed the BadEncoder attack, which injects a Trojan into the encoder during its training phase. Their aim is to ensure that all downstream classifiers, built on the Trojan-infected encoder and applied to different downstream tasks, simultaneously contain the Trojan. On the other hand, Feng et al. [17] introduced a detection strategy to identify such malicious pre-trained encoders. However, these works mainly focus on the integrity concern, while our work zeroes in on the privacy concern. Importantly, since the proposed ARO attack does not compromise model integrity, i.e., it does not contain any Trojan, it cannot be detected by Trojan detection methods [17].

Robustness overfitting. Recently, robustness overfitting has been identified as a significant issue in adversarial training (AT) [48]. This phenomenon has been partially explained through the lens of sample complexity [51], with empirical evidence drawn from related work, such as adversarial training using semi-supervised learning [2], [5], [69], robust local features [54], and data interpolation [32]. Although effective countermeasures exist to mitigate vanilla overfitting, these approaches prove insufficient in addressing robustness overfitting during adversarial training [48].

To address this problem, various strategies have been proposed, including the unequal treatment of data [70] and weight smoothing [9]. Recent research has also discovered a relationship between the flatness of the weight loss landscape and the robust generalization gap, providing valuable insights into the issue [63]. In addition, recent studies suggest that adversarial robust models are significantly more

susceptible to membership inference attacks [55]. This susceptibility can be partially explained: Robust models may reveal more membership information due to their larger generalization error in both benign and adversarial settings, i.e., leading to increased vanilla overfitting and robustness overfitting. However, to the best of our knowledge, no existing research has leveraged the actively induced robustness overfitting for advanced membership inference attacks, which forms the principal focus of our study.

8. Discussion

Limitations. The proposed ARO attack incurs a minor reduction in downstream accuracy compared to using a normal pre-trained model, as evaluated in Section 5.3 (shown in Table 2). For example, when the downstream dataset is CINIC-10, ARO_{β} achieves a downstream test accuracy of 70.62%, while using a normal pre-trained model can achieve an accuracy of 76.61%. However, we argue that this minor reduction is acceptable, since the victim may not know the level of accuracy that a normal pre-trained model could achieve. If the victim can achieve a higher test accuracy using a pre-trained model than by training from scratch, they might be inclined to use the pre-trained model, thus making the attack stealthy.

We should also mention that ARO_{β} represents only one specific implementation of the proposed ARO attack, and the attacker can manage the attack's stealthiness by adjusting the poisoning ratio. More comprehensive results can be found in Figure 5.

Future Work. ARO enhances membership inference attacks by leveraging a manipulated pre-trained model, thereby introducing a new privacy threat to the machine learning supply chain. Meanwhile, previous studies have demonstrated that manipulating the downstream dataset can further amplify membership inference attacks [10], [60]. Integrating these techniques to bolster overall attack performance remains an open research challenge, as does the development of effective countermeasures against such combined attacks.

9. Conclusion

In this study, we demonstrate the feasibility of exploiting pre-trained models to amplify the membership exposure in subsequent downstream tasks. In this paper, we presented ARO, a technique designed to create flawed pre-trained models that are detrimental to data privacy. ARO employs a distinctive training procedure incorporating input perturbation, weight perturbation, and data poisoning to actively induce robustness overfitting. Our exhaustive evaluation highlights that ARO is successful in amplifying membership leakage across numerous attack scenarios and multiple downstream tasks, while preserving the primary utility, i.e., downstream accuracy.

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Appendix A. Additional Figure

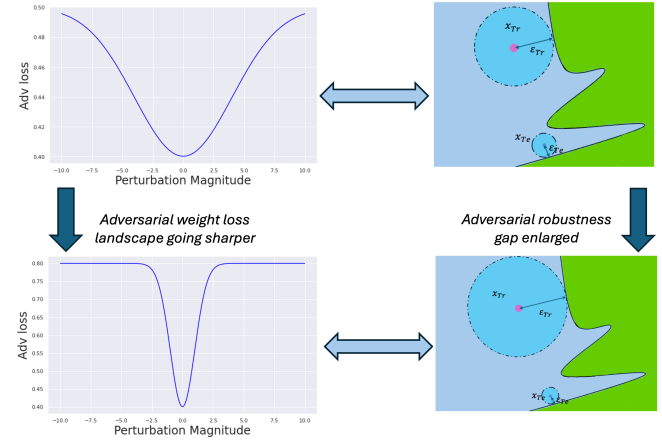


Figure 7: AWP works to mitigate robustness overfitting by flattening the adversarial weight loss landscape. For example, the figure illustrates two models with different adversarial weight loss landscape. The top half represents a model with a flat adversarial weight loss landscape, while the bottom half represents a model with a sharp adversarial weight loss landscape. The left side of the figure provides an intuition of the adversarial weight loss landscape, where the x-axis represents perturbations to the model parameters and the y-axis denotes the adversarial loss. The right side depicts how the adversarial robustness gap increases with a sharper adversarial weight loss landscape when considering two data samples: x_{Tr} , a training data sample, and x_{Te} , a testing data sample. This increase in gap is demonstrated through the robustness radii, ϵ_{Tr} for x_{Tr} and ϵ_{Te} for x_{Te} , highlighting the difference in their adversarial robustness.

Appendix B. Impact of Overfitting Perturbation Magnitude

As the input perturbation is part of the traditional adversarial training approach [40] and not unique to ARO, we fix the setting of the input perturbation part to better understand how overfitting perturbation influences ARO. In this experiment, we fix the downstream dataset as CelebA and ensure the upstream dataset is clean (without poisoning).

Observations. Figure 8 shows the variation in the level of downstream vanilla overfitting, the level of downstream robustness overfitting, and the attack AUC of MIAs when applying ARO with different γ values, which represent the magnitude of overfitting perturbation. We observe that the trend of effectiveness largely aligns with the trend of downstream robustness overfitting, while it demonstrates an inverse relationship with the level of downstream vanilla overfitting. Specifically, the Pearson’s correlation coefficient between vanilla overfitting and membership leakage is

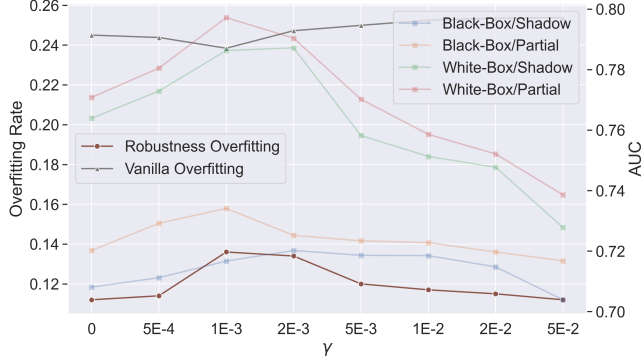


Figure 8: The change of vanilla overfitting rate, robustness overfitting rate, and AUC of MIAs with the changing perturbation magnitude γ .

−0.121, indicating a weak, inverse relationship. Conversely, the correlation coefficient between vanilla overfitting and robustness overfitting is 0.662, suggesting a moderate, positive linear relationship.

This finding indicates that robustness overfitting is the main factor through which ARO amplifies the membership leakage. Given that the trends of downstream vanilla overfitting and the membership leakage show inverse patterns with changes in γ , we can entirely dismiss the possibility that vanilla overfitting contributes to the enhancement of the effectiveness.

In addition, we observe that ARO has a clear upper limit when not supplemented with data poisoning. It achieves the highest AUC score when γ is around 1×10^{-3} . When γ exceeds 1×10^{-3} , both the robustness overfitting and the AUC score decrease. This could be because excessive overfitting perturbation impacts model convergence. Moreover, we observe that the upper bound of ARO is not sufficiently high (as per Table 1). Consequently, the use of data poisoning becomes somewhat necessary to facilitate a more aggressive attack.

Appendix C. Impact of Membership Inference Strategy

Setup. In this experiment, we evaluate two membership inference strategies: (i) $\mathcal{I}_A$: the inference strategy introduced by [55], which transforms all images into their adversarial examples (AE) subject to a perturbation constraint ϵ , and then uses these AE samples for inference. (ii) $\mathcal{I}_B$: a basic membership inference strategy that utilizes clean images for inference. Typically, $\mathcal{I}_B$ is equivalent to $\mathcal{I}_A$ with a perturbation constraint of $\epsilon = 0$. We consider four different l_∞ perturbation constraint ϵ values in this experiment: 0, 1×10^{-1} , 1×10^{-2} , 1×10^{-3} . The downstream dataset is fixed as CINIC-10 for analysis.

Observations. Table 4 illustrates the effectiveness of the proposed ARO_β using different membership inference strategy settings. We observe that ARO_β is more effective with

TABLE 4: Impact of Membership Inference Strategy

Perturbation Constraint ϵ	Black-Box\ Shadow	Black-Box\ Partial	White-Box\ Shadow	White-Box\ Partial
0 ($\mathcal{I}_B$)	0.8662	0.8876	0.9025	0.9129
1E-01 ($\mathcal{I}_A$)	0.7371	0.7509	0.7650	0.8001
1E-02 ($\mathcal{I}_A$)	0.8340	0.8686	0.8864	0.9030
1E-03 ($\mathcal{I}_A$)	0.8632	0.8882	0.8964	0.9080

$\epsilon = 0$, i.e., using $\mathcal{I}_A$, and generally, the smaller the ϵ , the better the effectiveness. This could be because the inference strategy proposed by [55] is optimized for models with extremely high robustness and may not be as effective against models with limited robustness. In our case, although we have applied adversarial training to the upstream model, inducing robustness overfitting, the model fine-tuned on downstream data tends to be less robust, particularly compared to a downstream model directly enhanced with adversarial training. This could be the reason why $\mathcal{I}_A$ demonstrates greater effectiveness in conjunction with ARO_β .

We note that although the best membership inference strategy remains the basic one, as opposed to querying with adversarial examples, we believe that adversarial robustness overfitting is a necessary approach for attacking transfer learning. This is because it is currently the only choice that can maintain stealth while amplifying membership leakage. As shown in Figure 5, inducing adversarial robustness overfitting significantly improves the effectiveness-stealthiness tradeoff.

Appendix D. The Algorithm

Algorithm 1 Active robustness overfitting

Input: Network after warm-up f_w , dataset $\mathcal{D}$, mini-batch $\mathcal{B}$, iteration number I , loss function ℓ , overfitting perturbation magnitude γ , PGD step size α , PGD steps K , PGD constraint ϵ .
Output: Adversarial robustness overfitted model f_w .

```

repeat
    Read mini-batch  $\mathcal{D}_B$  from training set  $\mathcal{D}$ .
    # Overfitting perturbation
     $\delta = \gamma \cdot \nabla_{\mathbf{w}}(\ell(f_{\mathbf{w}}(\mathcal{D}_B)))$ 
     $\mathbf{w} \leftarrow \mathbf{w} + \delta$ 
    # Adversarial training
    for all  $(x, y) \in \mathcal{D}_B$  do
        for  $k = 1$  to  $K$  do
             $x'_B \leftarrow \Pi_\epsilon(x'_B + \alpha \cdot \text{sign}(\nabla_{x'_B} \ell(f_{\mathbf{w}}(x'_B), y)))$ 
        end for
         $\mathbf{w} = \mathbf{w} - \alpha \nabla_{\mathbf{w}} \ell(x'_B, y; \mathbf{w})$ 
    end for
    # Revert perturbation
     $\mathbf{w} \leftarrow \mathbf{w} - \delta$ 
until repeat  $I$  times
return  $f_w$ 

```

Appendix E.

Meta-Review

The following meta-review was prepared by the program committee for the 2025 IEEE Symposium on Security and Privacy (S&P) as part of the review process as detailed in the call for papers.

E.1. Summary

This paper proposes to manipulate a pre-trained model to increase the privacy leakage, e.g., vulnerability to membership inference attacks, after fine-tuning. For this, it proposes an approach based on robust overfitting that transfers to downstream tasks. The methodology combines poisoning with adversarial training and adversarial weight perturbation techniques. The method is shown to enhance the performance of MIAs against fine-tuned models on multiple combinations of pre-trained and fine-tuned datasets. The effectiveness of the method is evaluated against several defense approaches, with differentially private training being the only effective defense.

E.2. Scientific Contributions

- Provides a Valuable Step Forward in an Established Field

E.3. Reasons for Acceptance

- 1) The paper provides a valuable step forward in an established field. Previous works have already explored the possibility of manipulating a pre-trained model in order to make a model fine-tuned on it more vulnerable to privacy attacks. The paper develops a new method that is not tailored to specific target records nor target datasets, and shows it to amplify membership leakage in the fine-tuned model without affecting downstream performance.

E.4. Noteworthy Concerns

- 1) The paper does not sufficiently compare with previous work to justify the novelty claims. The authors highlight in the rebuttal that compared to [43], [70], and [A], their attack does not require any knowledge of the downstream training data, and that compared with [A], it requires less complex model architecture modifications, increasing stealthiness. [A] Feng, S., & Tramèr, F. (2024). Privacy backdoors: stealing data with corrupted pretrained models. arXiv preprint arXiv:2404.00473.
- 2) The paper currently lacks insights about when and why robustness overfitting transfers from pre-trained models (e.g., impact of the similarity between distributions of pre-trained and fine-tuned datasets on the attack performance).
- 3) Claims of statistical significance are not supported with error bars or statistical tests.